

Mukund Telukunta

<https://mukundtelukunta.com>
github.com/mukund0911

(952) 649-2653
mukundtelukunta@gmail.com
[linkedin.com/in/mukundtelukunta](https://www.linkedin.com/in/mukundtelukunta)

SUMMARY

Applied scientist with a PhD and 5+ years across the full ML model lifecycle: research, NLP and GenAI model building, agentic AI systems, production deployment, and rigorous evaluation. Built WaveCrafter: LangGraph agentic AI system with multi-model routing, RAG, and rigorous agent evaluation (composite 6-metric LLM-as-Judge, offline regression suites); **webhook p99 <200ms, 90% cost reduction**. Pretrained DNABART (103M-param BART) from scratch; **SOTA on 16/26 benchmark tasks, 24% over a 2.5B baseline**. First author across four peer-reviewed venues (CHIL 2024 PMLR, ECML PKDD 2023, RecSys 2020, arXiv 2025); strong Python and PyTorch.

EDUCATION

Missouri University of Science and Technology

PhD in Computer Science; M.S. Computer Science (Dec 2020)

Rolla, MO

May 2025

Focus: NLP, agentic AI systems, generative model evaluation, applied ML research

Awards: 1st Place Graduate Research Poster (2019); Runner-up IEEE St. Louis Student Presentations (2022); Invited Panelist (May 2024, Nov 2024).

Publications: CHIL 2024 PMLR 248:683-695 (first author); ECML PKDD 2023 (first author); IEEE MASS 2023; RecSys 2020 (first author); arXiv:2505.04886 (under review, first author).

TECHNICAL SKILLS

NLP & Generative AI: BART/transformer pretraining from scratch, LLMs, RAG, prompt engineering, sequence-to-sequence generation, embeddings, information retrieval, HuggingFace Transformers

Agentic AI & Evaluation: LangGraph, multi-agent orchestration, tool calling, multi-model routing, LLM-as-Judge, agent evaluation, 26-task benchmarks, drift monitoring, Anthropic Claude API, OpenAI API

ML & Deep Learning: PyTorch, TensorFlow, supervised/unsupervised learning, computer vision, GNNs, causal inference, Bayesian inference, DDP + NCCL, mixed-precision, fine-tuning, explainability

MLOps & Production: Python, SQL, MLflow, SageMaker, W&B, FastAPI, Docker, Kubernetes, GitHub Actions CI/CD, AWS, Modal.com, Kafka + Spark, Prometheus + ELK

PUBLICATIONS (<https://scholar.google.com/citations?user=iNSSYw8AAAAJ>)

1. **Telukunta, M.**, & Nadendla, V. S. S. "A Difficulty in Evaluating Perceived Group Fairness using Non-Expert Opinions." (In preparation.)
2. **Telukunta, M.**, Stuart, M., Nadendla, V. S. S., & Canfield, C. "Fairness Perception of Regression-based Predictive Analytics Tools used in Kidney Placement." (In preparation.)
3. **Telukunta, M.**, Rao, S., Stickney, G., Nadendla, V.S.S. & Canfield, C. (2024). "Learning Social Fairness Preferences from Non-Expert Stakeholder Opinions in Kidney Placement." *Proceedings of the fifth Conference on Health, Inference, and Learning (CHIL)*, in *Proceedings of Machine Learning Research*.
4. Sanga, S., **Telukunta, M.**, Nadendla, V. S. S., & Das, S. K. (2023). "Strategic Information Design in Selfish Routing with Quantum Response Travelers." *IEEE 20th International Conference on Mobile Ad Hoc Smart Systems (MASS)*.
5. **Telukunta, M.**, & Nadendla, V. S. S. (2023) "Towards Inclusive Fairness Evaluation via Eliciting Disagreement Feedback from Non-Expert Stakeholders." *3rd Workshop on Bias and Fairness in AI at European Conference on Machine Learning (ECML PKDD)*.
6. **Telukunta, M.**, & Nadendla, V. S. S. (2020) "On the Identification of Fair Auditors to Evaluate Recommender Systems based on a Novel Non-Comparative Fairness Notion." *3rd FAccTRec Workshop on Responsible Computing at ACM Recommender Systems (RecSys)*.

RESEARCH EXPERIENCE

Applied Scientist / ML Researcher

August 2020–May 2025

Missouri University of Science and Technology

Rolla, MO

- Designed, built, and rigorously evaluated an agentic AI system (WaveCrafter): LangGraph multi-agent orchestration (3 production graphs, 4 named agents, Anthropic Claude and OpenAI multi-model routing, Pydantic tool calling, RAG over PostgreSQL); rigorous agent evaluation via composite 6-metric LLM-as-Judge scorer, offline golden-set regression suites, and online drift monitoring; **webhook p99 <200ms, 90% cost reduction, 3× latency reduction**.
- Pretrained DNABART (103M-param BART encoder-decoder) from scratch: BPE tokenizer (4,096-token vocabulary), 3 self-supervised pretraining objectives, DDP with NCCL on H100/V100 (batch=128, 4 days, mixed-precision); **SOTA on 16/26 GUE benchmark tasks, 24% improvement over 2.5B Nucleotide Transformer at 4% of parameters**; evaluated across 26 downstream NLP tasks; full wandb.watch instrumentation.

- Owned the full model lifecycle end-to-end: model selection, training, fine-tuning, explainability (attention rollout, ablation studies), and evaluation (26-task benchmark, failure-mode analysis); productionized models with MLflow champion-challenger shadow deployment, drift monitoring, and auto-rollback – the model-selection-through-deployment-through-evaluation cycle the Core AI team runs.
- Conducted rigorous applied research validated through 15+ experiments (CHIL 2024 PMLR, first author): A/B testing, propensity matching, difference-in-differences, switchback; Mixed-Logit + Bayesian modeling over N=5,000+ human judgments; collaborated with domain-expert stakeholders (transplant surgeons) to translate requirements into deployed systems; published 4 peer-reviewed papers.
- Built supervised, unsupervised, and generative models across modalities: NLP (DNABART), computer vision (PitchVision: YOLO, ResNet, DeepSORT, **<20ms on NVIDIA Jetson**), generative audio (330M-param, **90% cost reduction**), graph representation learning (HGT, **R² 0.91**), and RL-adjacent optimization (IEEE MASS 2023, **50% cost reduction**).
- Communicated technical findings to non-technical stakeholders and cross-functional teams; led design reviews and modeling discussions; mentored 2 junior researchers and 3 interns; presented research as an invited panelist.

ML Research Collaborator

October 2024–January 2025

Bioinformatics Research Lab, Missouri University of Science and Technology

Rolla, MO

- Productionized a 150M-param model from research notebooks to production serving: Docker on Kubernetes, MLflow versioning, champion-challenger shadow deployment with auto-rollback, C extension tokenizer (**5× speedup**); drift monitoring (JS divergence, confidence thresholds).
- Built pretraining data pipeline: Kafka + Spark Streaming + Parquet on S3, Airflow batch retraining; DDP on 4× A100 (bf16, gradient checkpointing); **5× more data in 33% less wall-clock time**.

Software Engineer (Internship)

June–August 2019

AT&T Labs

Des Peres, MO

- Built cloud analytics platform on AWS and GCP with automated ETL pipelines and real-time dashboards; **30% forecasting improvement, 95% adoption across 800-person org**.

SELECTED PROJECTS ([GITHUB.COM/MUKUND0911](https://github.com/mukund0911))

WaveCrafter: Agentic AI System with Rigorous Agent Evaluation | wave-crafter.com

- LangGraph multi-agent orchestration (3 production graphs, 4 named agents, Anthropic Claude + OpenAI multi-model routing selecting per-task by cost and quality, Pydantic tool calling, RAG over PostgreSQL with timestamped citation, SSE streaming); agents reason through multi-step workflows, execute tool calls, and recover from failures autonomously.
- Rigorous agent evaluation: composite 6-metric LLM-as-Judge scorer auto-rejecting low-quality outputs before user exposure, offline golden-set regression suites replaying before every model or prompt change, online drift monitoring, human review queues feeding labeled signal back into improvement; **webhook p99 <200ms, 90% cost reduction (\$350 to \$33/month), 3× latency reduction**; zero-downtime GitHub Actions CI/CD across 3 services.

DNABART: NLP Foundation Model — Pretraining and 26-Task Evaluation |

github.com/mukund0911/dnabart

- 103M-param BART encoder-decoder pretrained from scratch: BPE tokenizer, 3 self-supervised objectives, DDP + NCCL on H100/V100; **SOTA on 16/26 tasks, 24% over 2.5B baseline at 4% of parameters**.
- 26-task evaluation framework; ablation studies; attention-rollout explainability; failure-mode analysis; wandb.watch full instrumentation.

PitchVision: Multi-Modal Perception with LLM Query Layer | github.com/mukund0911/pitch-vision

- Built every neural network from scratch in PyTorch: ResNet-18 backbone + anchor-based YOLO detection head (**87.3% mAP@0.5, 42 FPS**), DeepSORT tracker (8D Kalman filter from scratch, 128-dim ReID embeddings, Hungarian assignment; **MOTA 71.4, IDF1 68.9**), 2M-param pre-norm Transformer for trajectory prediction trained via imitation learning (**1.4m MAE, 78.6% intent accuracy**).
- **<20ms end-to-end GPU inference on NVIDIA Jetson** with C++ performance-critical path; PyTorch Profiler-driven optimization; FastAPI inference server with LLM agent query layer routing natural-language queries over a structured match event log.

Heterogeneous Graph Transformer: Representation Learning at 700K-Node Scale

- HGT over a 700K-node, 1.8M-edge graph (12 relation types, 5 message-passing hops, node-type-specific K/Q/V attention) with GraphSAGE + LSTM inductive inference generalizing to novel entities without retraining, and a Space2Vec Fourier-basis spatial encoder; benchmarked 4 GNN variants (GCN, GraphSAGE, GAT, HGT) in a unified evaluation framework.
- **MAE 0.047, R² 0.91, 23% lower error vs. GCN baseline, 60% reduction in manual assessment effort**; REST API serving real-time predictions at sub-100ms latency; mini-batch neighbor sampling reducing per-epoch training **35%**; UMAP cosine >0.82 confirming embedding quality.